Hardware Bring-Up Technical Report

NMS — EVT 1 Custom PCB

PALROC

Prepared for: Nova Mobile Systems
Cody Sharff

Prepared by: PALROC 2025 SL
Oriol Reverte Roca
NIF: B26945105
oriol@palroc.com
+34 633 154 834

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Executive summary

This report documents the hardware bring-up of the NMS EVT 1 custom PCB, a compact IoT node combining cellular connectivity, satellite positioning, Bluetooth Low Energy, Wi-Fi scanning, and a battery management system on a single board.

The bring-up was carried out incrementally across four physical boards using Nordic Semiconductor's nRF Connect SDK (NCS) v3.3.0 and a SEGGER J-Link for SWD programming and RTT logging. Every primary chip on the board has been individually verified as electrically alive and functional on its bus. The LTE-M/NB-IoT, BLE, and Wi-Fi radio stacks have been validated end-to-end. GNSS will be validated in EVT 2 after the LNA supply path is corrected (see Section 4).

The board is ready to hand off to application firmware development. A complete set of bring-up artefacts — per-peripheral probe firmware, a structured bring-up log, a per-board test runner, and reusable lessons — is included with this report.

Operational caveat: bring-up validates each chip individually. Because the nRF9151 and nRF54L15 share the I2C2 and SPI3 buses, the two MCUs cannot run their bring-up firmware simultaneously — the L15 flash must be erased (or the L15 held in reset) while the 9151 side is being tested, and vice versa. Designing a software bus arbitration protocol between the two MCUs is delegated to application firmware development. The same applies to the inter-MCU UART link, which is wired and powered but not yet functional under firmware (root cause and fix recommendation provided).

Item	Detail
Board	NMS EVT 1 (PALROC design)
SDK	nRF Connect SDK v3.3.0 / Zephyr RTOS 4.3.99
Chips validated	nRF9151, nRF54L15, nRF7000, nPM1300
Boards tested	4 units (boards #2, #3, #4, #5); board #1 destroyed
Radios validated	LTE-M/NB-IoT ✓ BLE ✓ Wi-Fi ✓ GNSS: EVT 2
Status	**Bring-up complete. Ready for application firmware.**

Platform overview

Board architecture

Chip	Function	Interface	Supply
Nordic nRF9151 SiP	LTE-M/NB-IoT + GNSS	Primary MCU	VBAT
Nordic nRF54L15	BLE / Thread / Matter	SWD; UART (EVT 2)	BUCK2
Nordic nRF7000	Wi-Fi 6 scan-only	SPI1 (dedicated)	BUCK1 + BUCK2
Nordic nPM1300	PMIC: BUCKs, LDOs, charger	I2C2	VSYS
Winbond W25Q128JV	16 MB NOR flash	SPI3 (shared)	BUCK2
ST LSM6DSO	IMU (accel + gyro)	SPI3 → I2C (EVT 2)	BUCK2
GNSS LNA	Antenna amplifier	COEX0 gate	LDO1
BGS12SN6E6	2.4 GHz antenna SPDT	GPIO SW_CTRL0	LDO2
Hall sensors ×2	Magnetic proximity	GPIO P0.00, P0.01	—
LEDs ×3	Status indicators	GPIO + MOSFET	BUCK2

Supply topology

Rail	Voltage	Consumers
VSYS	Battery / MPPT / VBUS	nPM1300 input
BUCK1	3.3 V	nRF7000 VBAT
BUCK2	3.3 V	nRF7000 VDDIO, W25Q128JV, LSM6DSO, nRF54L15, LEDs
LDO1	3.3 V regulated	GNSS LNA VDD (damaged in EVT 1 — see Section 4)
LDO2	off	BGS12SN6E6 RF switch VDD (not needed in EVT 1)

Validation results

Board history

Four boards were used during EVT 1:

- **Board #1 — destroyed.** A Wi-Fi driver misconfiguration triggered a fatal error in the nPM1300, permanently damaging the board. This board is not counted in the validation results.
- **Boards #2, #3, #4, #5 — all validated.** All four show identical behaviour across every test. Results reported below apply to all four.

Per-subsystem summary

Subsystem	Result	Notes
nRF9151 boot + RTT	PASS	Stable on all 4 boards
LTE-M / NB-IoT attach	PASS	NB-IoT B20 Orange ES roaming; 76 s attach on board #2
GNSS receive path	FAIL	LNA burned by 5 V overvoltage; deferred to EVT 2
nPM1300 telemetry	PASS	VBAT, NTC temp, IBAT readable via I2C2
Battery charging	PASS	158 mA confirmed (150 mA target)
BUCK1 / BUCK2 rails	PASS	Both at 3.3 V, multimeter-verified
LDO1 / LDO2 (regulated)	PASS	LDO mode confirmed; 3.3 V regulated output
I2C2 bus	WARN	Works only when L15 flash is erased; fails with both MCUs active
SPI3 flash (W25Q128JV)	FAIL	Damaged by 5 V overvoltage (same root cause as LNA)
SPI3 IMU (LSM6DSO)	FAIL	Never validated; moving to I2C in EVT 2
nRF7000 Wi-Fi scan	PASS	9 APs detected; passive scan enforced
nRF54L15 boot + RTT	PASS	After K32SRC Kconfig fix + nRESET pull-up rework
nRF54L15 BLE advertising	PASS	Validated on phone BLE scanner at 15 m through walls
nRF54L15 ↔ nRF9151 UART	FAIL	Tested; `lpuart` REQ/RDY handshake works but zero data bytes received — root cause: deprecated `UART_0_NRF_HW_ASYNC` Kconfig in NCS 3.3.0; fix is DT `timer = <&timer2>` phandle. Delegated to firmware engineer
Hall sensor hall2 (DRV5032FBDBZR, P0.01)	PASS	Push-pull, no magnet = HIGH, magnet = LOW; direct-connect to SHPHLD validated
Hall sensor hall1 (P0.00)	FAIL	Broken on all boards; footprint/supply under investigation
LEDs ×3	PASS	After MOSFET package rework
MPPT input	PASS	Wrong passive values in v1 set MPPT threshold too low; corrected resistor BOM, MPPT now triggers correctly (bench-validated)

Per-board test runner results

A lightweight per-board test runner (`test_report()` / `test_report_summary()`) was integrated into both the nRF9151 and nRF54L15 firmware. Results are identical across boards #2–#5.

Test	Board #2	Board #3	Board #4	Board #5
modem_init	PASS	PASS	PASS	PASS
lte_attach	PASS	PASS	PASS	PASS
npm1300	PASS	PASS	PASS	PASS
spi_flash	FAIL	FAIL	FAIL	FAIL
spi_imu	FAIL	FAIL	FAIL	FAIL
wifi	PASS	PASS	PASS	PASS
L15 boot	PASS	PASS	PASS	PASS
L15 BLE advertise	PASS	PASS	PASS	PASS

Key technical findings

Root cause: LDO load-switch mode and 5 V overvoltage

The most significant finding of EVT 1 is a single root cause responsible for multiple component failures: the nPM1300 LDO channels (LDO1, LDO2) were in load-switch mode by default, with their input (LDOIN) tied directly to VSYS. In load-switch mode the LDO output tracks the input — it does not regulate. When a USB VBUS (5 V) source was connected, VSYS rose to 5 V and both LDO outputs followed, exposing all downstream components to 5 V regardless of their rated supply voltage.

Confirmed damage from this event:

- **GNSS LNA** (max VCC = 3.4 V): exposed to 5 V, burned. GNSS receive path non-functional.
- **W25Q128JV NOR flash** (on BUCK2 rail): also exposed; registers corrupt reads. Board #1 showed a working flash read before this event; no board recovered afterwards.
- **3.3 V rail** in general: voltage spiked to approximately 4.2 V during the event.

Fix applied in EVT 1 firmware: Switched both LDOs to regulated LDO mode via DTS (`NPM13XX_LDSW_MODE_LDO`). In LDO mode, the output is regulated to the configured voltage regardless of LDOIN. LDO1 and LDO2 now output 3.3 V stably.

v2 fix: Connect LDOIN to the regulated 3.3 V BUCK output (not VSYS), and add 20 µF output capacitors required for LDO stability. This permanently caps the LDO output at 3.3 V regardless of VBUS.

GNSS deferred to EVT 2

The GNSS LNA was burned by the 5 V overvoltage event described above. GNSS tracking was never established on EVT 1 hardware. The satellite receive path will be validated on EVT 2 boards once the LNA supply is corrected and the LNA is replaced.

SPI3 flash and IMU

The W25Q128JV flash was seen returning a correct JEDEC ID on board #1 before the overvoltage event. After the event, all boards return corrupt data. The LSM6DSO IMU was never seen working on any board. Both devices will be addressed in EVT 2:

- The LSM6DSO will move from SPI3 to I2C, which simplifies the bus topology and eliminates the dual-master contention risk.
- The nRF54L15 and nRF9151 will communicate over UART rather than sharing SPI3. This avoids any bus arbitration complexity between the two MCUs (a recommendation from Nordic's own developer forum). UART inter-MCU communication is the next item to validate.

I2C2 bus sharing — not validated with both MCUs active

The nRF9151 and nRF54L15 share the I2C2 bus. When both MCUs are running firmware, I2C2 traffic from the 9151 (e.g. nPM1300 register accesses) is corrupted — the bus does not work reliably with both chips active. The 9151 side works correctly only when the L15 flash is erased (or the L15 is held in reset).

An L15-side `SYS_INIT` hook was added to park P1.4 (SDA) and P1.5 (SCL) as high-Z inputs before any drivers start, with the intention of preventing the L15 from driving the shared lines.

This was not sufficient to make I2C2 work with both MCUs active — the bus still fails when the L15 firmware is running.

Resolving this is delegated to the application firmware engineer. A bus arbitration protocol between the two MCUs (likely over the inter-MCU UART, once that path is validated) is required before both sides can run simultaneously without bus contention. Until that protocol exists, the operational rule is: **erase the L15 before testing the 9151 side, and vice versa**.

nRF54L15 without LFXO crystal

The nRF54L15 on this board has no 32 kHz crystal. Without a specific Kconfig pair (`CONFIG_CLOCK_CONTROL_NRF` and `CONFIG_CLOCK_CONTROL_NRF_K32SRC_XTAL=n`), the Zephyr kernel hangs at boot waiting for a clock source that never arrives. Disabling the LFXO in the devicetree alone is insufficient; both Kconfigs are required.

nRF9151 modem firmware

The nRF9151 ships with PTI protocol-testing firmware only — not the modem firmware required for LTE operation. Before any modem work, the modem firmware (`mfw_nrf91x1`) must be flashed using the **nRF Connect for Desktop** programmer application (via the "Program LTE modem firmware" flow). Attempting to initialise the modem without this step results in a hard crash (`PC=0`, error `0xfff`).

nRF7000 Wi-Fi: sysbuild Kconfig only

In NCS v3.3.0 the nRF7000 is configured exclusively through sysbuild Kconfigs (`sysbuild.conf`). Setting any `CONFIG_WIFI_NRF70*` flag in `prj.conf` has no effect and causes CMake errors. Correct flags: `SB_CONFIG_WIFI_NRF70=y`, `SB_CONFIG_WIFI_NRF70_SCAN_ONLY=y`, `SB_CONFIG_WIFI_PATCHES_EXT_FLAS`

Hall sensor hall1 broken

Hall sensor 1 (P0.00) is non-functional on all boards tested. Hall sensor 2 (P0.01) works correctly as an active-LOW open-drain input. The hall1 footprint or supply path should be investigated before EVT 2.

EVT 2 hardware recommendations

#	Change	Reason
1	Connect LDOIN to regulated 3.3 V BUCK	Prevents 5 V overvoltage on VBUS; root cause of LNA + flash damage
3	Verify / replace GNSS LNA	Burned by overvoltage (Vmax = 3.4 V, exposed to 5 V)
4	Move LSM6DSO from SPI3 to I2C	Simpler topology; eliminates dual-master risk
6	Inter-MCU: UART instead of shared SPI	Eliminates bus contention; validated pattern per Nordic DevZone
7	Tie VPV to GND when no PV panel	Floating MPPT input caused anomalous rail behaviour
8	Remove Q4; connect Hall sensor (DRV5032FBDBZR) directly to SHPHLD	Q4 inverter made polarity backwards; original circuit reset nPM1300 every 10 s without magnet. Direct connection works (bench-validated)
9	Investigate hall1 footprint/supply	Broken on all 4 boards
10	TVS diode on VBUS rail	Transient protection on USB input; protects downstream from VBUS surge events

Delivered artefacts

- **Bring-up log (`steps.md`):** 1600+ lines of fully attributed bring-up history. Every Kconfig choice, devicetree decision, and rework is documented with the reasoning behind it. A new firmware engineer can read it cold and understand the full EVT 1 history.
- **Per-peripheral probe firmware (`src/`):** Individual probe functions for the nRF9151, nPM1300, W25Q128JV, LSM6DSO, nRF7000, hall sensors, and LEDs. Serves as reference code and starting templates for application development.
- **nRF54L15 BLE broadcaster app (`nrf54l15/`):** Validated BLE advertising on the custom board, including LFXO Kconfig fix, shared I2C pin parking, and nRESET handling.
- **Test runner (`src/test_report.c/h`):** Lightweight PASS/FAIL/SKIP/INFO per-test recorder with a structured summary table printed via RTT on every boot. Integrated into both the 9151 and L15 firmware.
- **Per-board test summary (`test_summary.md`):** Captured results for boards #2–#5 with a cross-board comparison table.
- **Custom board files (`boards/palroc/`):** Complete Zephyr board definitions for `nrf9151_palroc/nrf9151/` and `nrf54l15_palroc` with correct devicetree, pinctrl, and Kconfig.
- **EVT 2 hardware delta list:** 12 evidence-driven recommendations as listed in Section 5.

Sign-off

This report confirms that the NMS EVT 1 hardware bring-up is complete. LTE-M/NB-IoT, BLE, and Wi-Fi have been validated end-to-end. GNSS and the SPI peripherals are deferred to EVT 2 pending hardware corrections. The platform is ready for application firmware development.

Acknowledged by

Signature: __________________

Name: __________________

Title: __________________

Date: __________________

Prepared by

Oriol

Name: Oriol Reverte Roca

Title: Hardware Engineer

Date: May 5, 2026

PALROC 2025 SL
NIF: B26945105

Contact: Oriol Reverte Roca
PALROC 2025 SL NIF: B26945105
oriol@palroc.com
+34 633 154 834

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